

Motion and Forces

Subject: Physics

Grade: Grade 9

Learn how speed, velocity, and forces describe object movement and change.

Speed and Velocity

Speed tells how fast something moves. Velocity also includes direction, so it describes motion more completely.

Acceleration

Acceleration is the change in velocity over time. It happens when an object speeds up, slows down, or changes direction.

Forces and Motion

Forces like pushes and pulls change motion. Balanced forces keep motion steady, while unbalanced forces cause acceleration.

Practical Examples

Objects on a bicycle, a ball rolling down a slope, and a car braking all show how motion and forces work in real life.